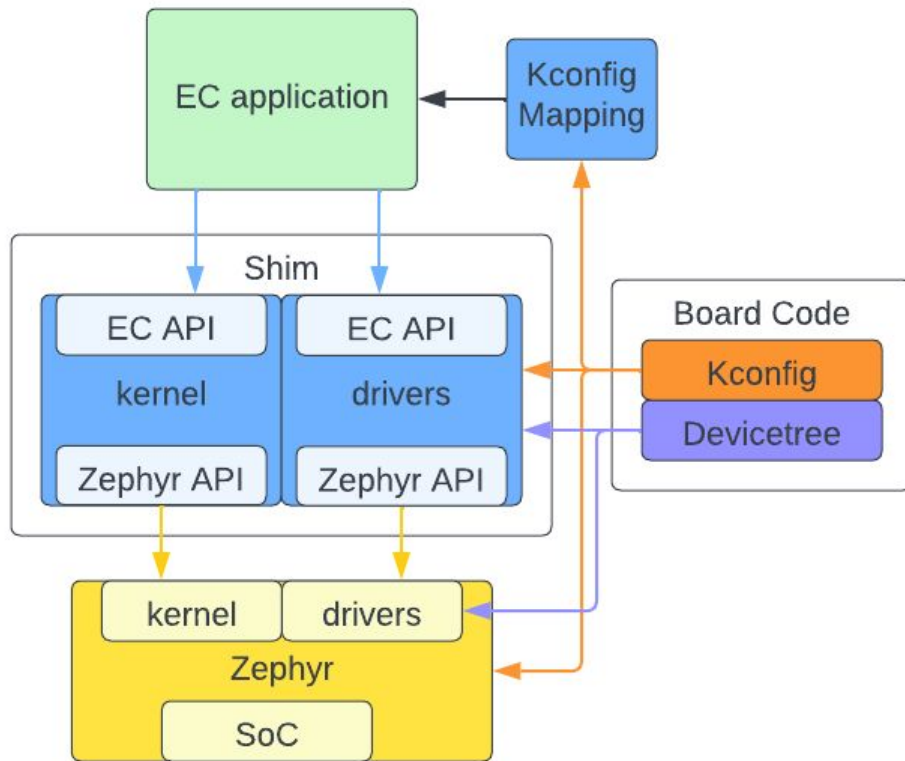


Migrating existing code to Zephyr

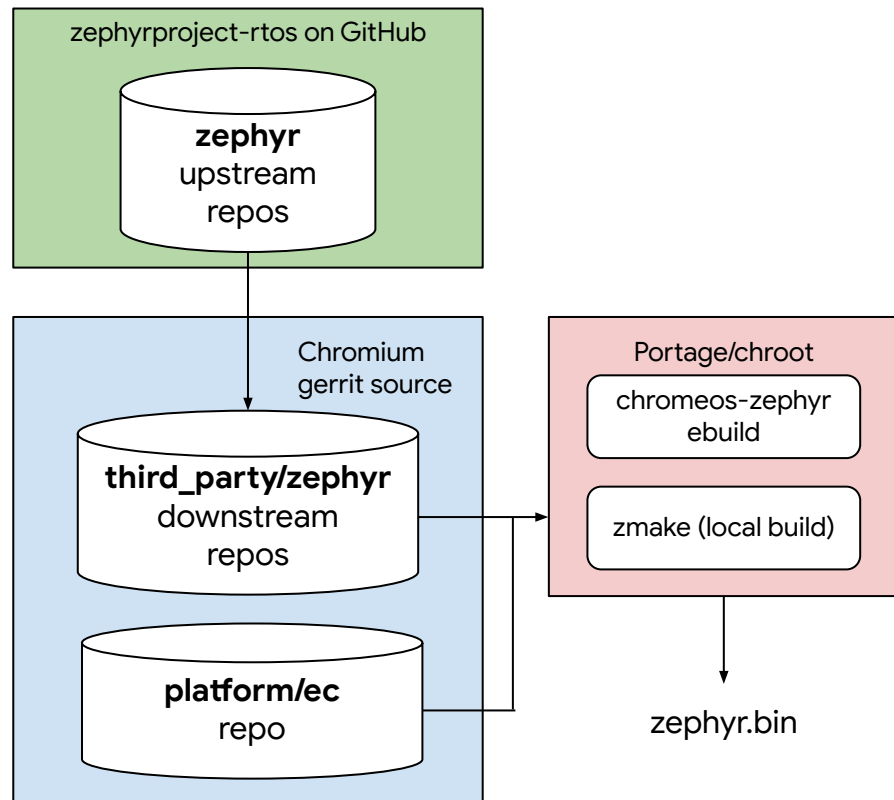
- Shim layer
 - Translate existing APIs to Zephyr APIs
 - Configured via Kconfig and devicetree
 - Reduces custom board code
- Kconfig mapping
 - Maps Kconfig options to existing `#defines`



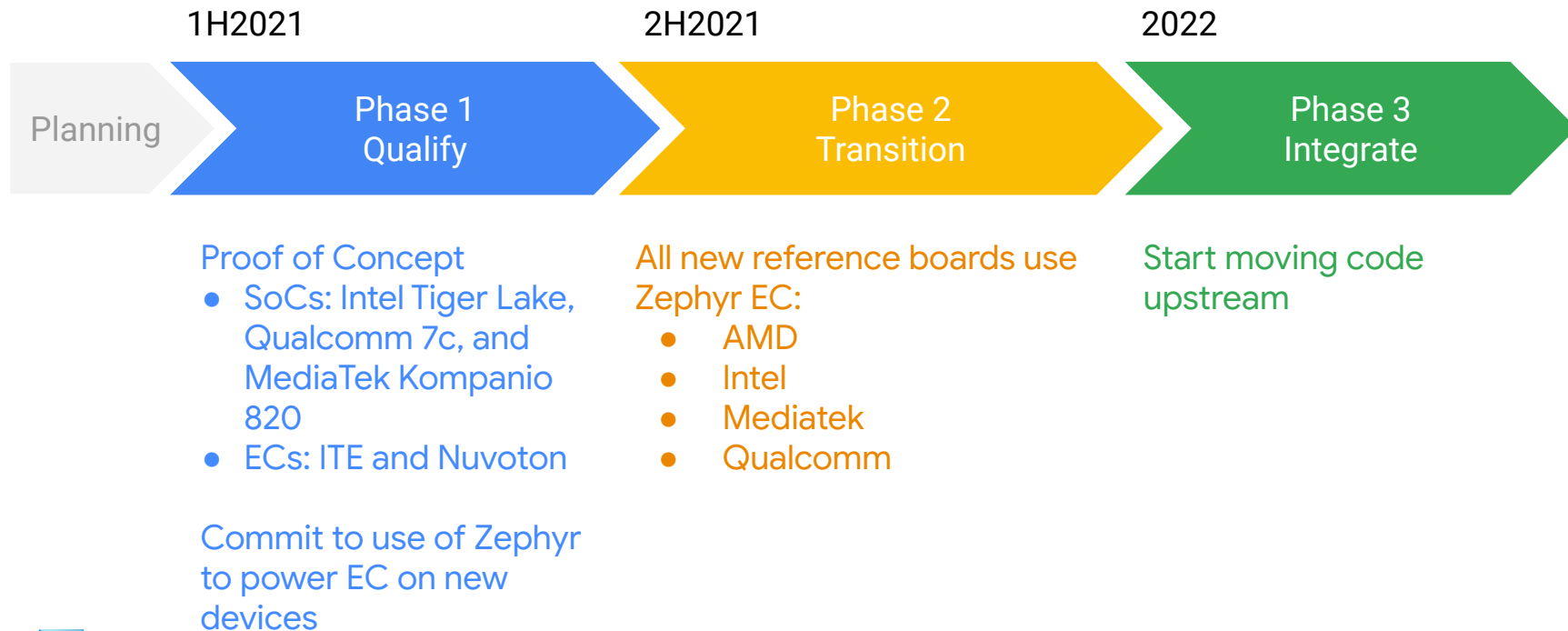
Zephyr EC structure

Multiple Chromium repositories

- [third_party/zephyr](#)
 - mirror of [Zephyr upstream on GitHub](#)
- [platform/ec](#)
 - Shared code with legacy EC
 - Zephyr shim and board projects under [platform/ec/zephyr](#) directory



Timeline



Hardware-free testing

- Tests use public APIs for verification
- There are 2 types of tests
 - Driver tests
 - Integration tests
- Emulators
 - [I2C controller](#)
 - I2C targets
 - USB-C Power delivery partner
- Emulators are in platform/ec/zephyr/emul and tests are in platform/ec/zephyr/test
- Coverage reports are generated in our [GitLab CI](#), and are publically available



References

- Google Security Chip: <https://showcase.withgoogle.com/titan-c/>
- Google's original EC code introduction:
<https://chromium.googlesource.com/chromiumos/platform/ec/+/HEAD/README.md>
- Zephyr EC Application
 - [Introduction Document](https://chromium.googlesource.com/chromiumos/platform/ec/+/HEAD/docs/zephyr/README.md):
<https://chromium.googlesource.com/chromiumos/platform/ec/+/HEAD/docs/zephyr/README.md>
 - [Source code](https://source.chromium.org/chromiumos/chromiumos/codesearch/+/main:src/platform/ec/):
<https://source.chromium.org/chromiumos/chromiumos/codesearch/+/main:src/platform/ec/>
- Upstream code
 - [State Machine Framework](https://docs.zephyrproject.org/latest/services/smf/index.html): <https://docs.zephyrproject.org/latest/services/smf/index.html>
 - [USB-C Drivers](https://github.com/zephyrproject-rtos/zephyr/tree/main/drivers/usbc): <https://github.com/zephyrproject-rtos/zephyr/tree/main/drivers/usbc>
 - [CHRE module](https://github.com/zephyrproject-rtos/chre): <https://github.com/zephyrproject-rtos/chre>
 - I2C Controller Emulator:
https://github.com/zephyrproject-rtos/zephyr/blob/main/drivers/i2c/i2c_emul.c



References (continued)

- Test Coverage: <https://gitlab.com/zephyr-ec/ec/-/jobs>
- Chromium repositories
 - third_party/zephyr:
https://source.chromium.org/chromiumos/chromiumos/codesearch/+/-/main:src/third_party/zephyr/
 - platform/ec:
<https://source.chromium.org/chromiumos/chromiumos/codesearch/+/-/main:src/platform/ec/>
- Closed case debug (CCD)
 - https://chromium.googlesource.com/chromiumos/third_party/hdctools/+/-/HEAD/docs/ccd.md
-

Zephyr EC threads (highest to lowest priority)

- USB-C Power delivery interrupt (1 per type-C port)
- USB-C Power delivery thread (1 per type-C port)
- Keyboard scanning
- AP command processing
- Sensors
- AP Power Sequencing
- Shell
- Charger
- BC1.2 host/client (1 per type-C port)
- System Work Queue
- Idle

